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EXP NO: 1 DATE:	AZURE DEVOPS ENVIRONMENT SETUP
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AIM:-

To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

INSTALLATION:-

1.Open your web browser and go to the Azure website:

<https://azure.microsoft.com/en-us/get-started/azure-portal>.

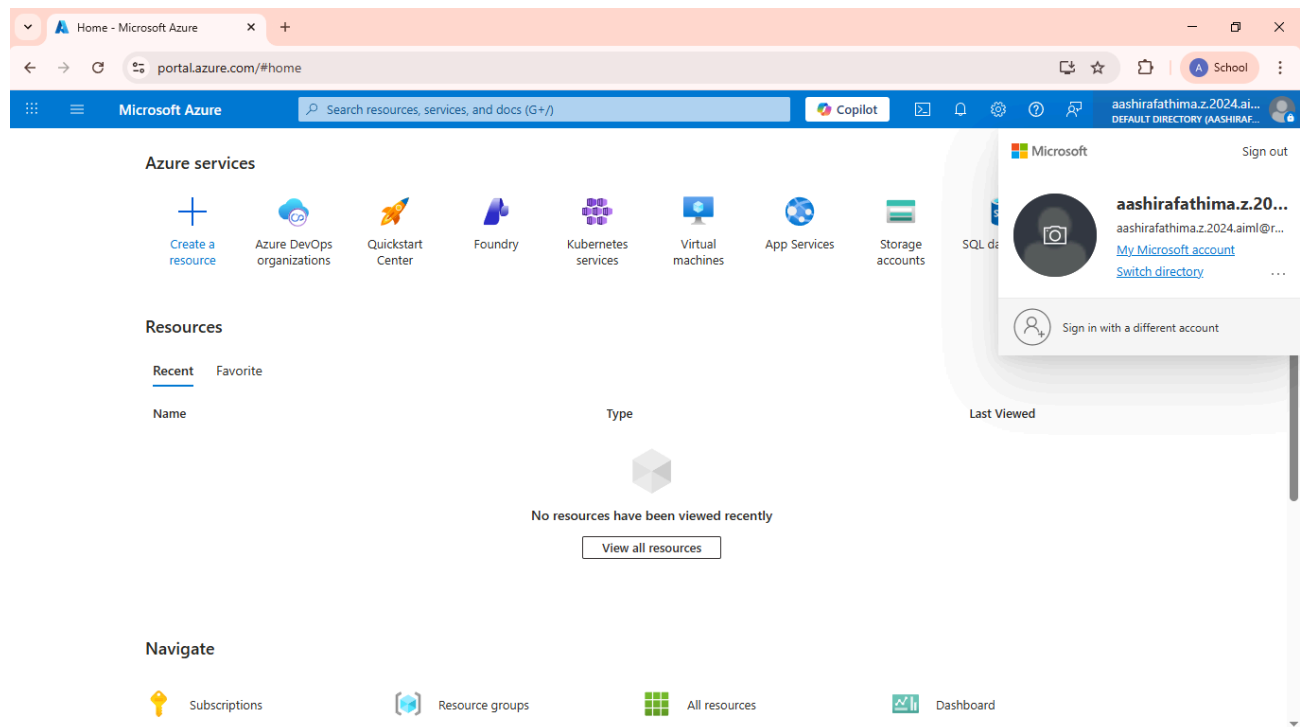
Sign in using your Microsoft account credentials.

If you don't have a Microsoft account, you can create one here:

<https://signup.live.com/?lic=12116231501146CS23432>

2.Azure home page

3.Open DevOps environment in the Azure platform by typing *Azure DevOps Organizations* in the search bar.



4. Click on the ***My Azure DevOps Organization*** link and create an organization and you should be taken to the Azure DevOps Organization Home page.

RESULT:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

EXP NO: 2 DATE:	AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT
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AIM:-

To set up an Azure DevOps project for efficient collaboration and agile work management.

- 1.Create An Azure Account
- 2.Create the First Project in Your Organization
 - a) After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
 - b) On the organization's **Home page**, click on the **New Project** button.
 - c) Enter the project name, description, and visibility options:
 - Name:** Choose a name for the project (e.g., **LMS**).
 - Description:** Optionally, add a description to provide more context about the project.
 - Visibility:** Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
 - d) Once you've filled out the details, click **Create** to set up your first project.
- 3.Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.
- 4.Project dashboard
- 5.To manage user stories:
 - a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.
 - b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

Azure DevOps aashirafathimaz2024aiml / STUDENT DATABASE MANA... / Boards / Backlogs

STUDENT DATABASE MANAGEMENT Team

Backlog Analytics

Filter by keyword Types Assigned to States Area Iteration Tags

	Order	Work Item Type	Title	State	Effort	Busin...	Value Area	Tags
	1	Epic	👑 Login and Authentication	New			Business	
		Feature	👑 Student and Teacher login	New			Business	
		User Story	👤 As a student or teacher I want to enter my username a...	New			Business	
		User Story	👤 As an admin I want to edit student Profiles	New			Business	
		User Story	👤 As an admin I want to delete student profiles	New			Business	
	2	Epic	👑 Academic details Management	New			Business	
		Feature	👑 Academic data validation	New			Business	
	3	Epic	👑 Batch import and data upload	New			Business	
		Feature	👑 Batch student data upload	New			Business	
		Feature	👑 Data validation and error handelling	New			Business	
	4	Epic	👑 Student Data Access and Reporting	New			Business	
		Feature	👑 Student data search	New			Business	
		Feature	👑 student data search	New			Business	

Result

Successfully created an Azure DevOps project with user story management and agile workflow setup.

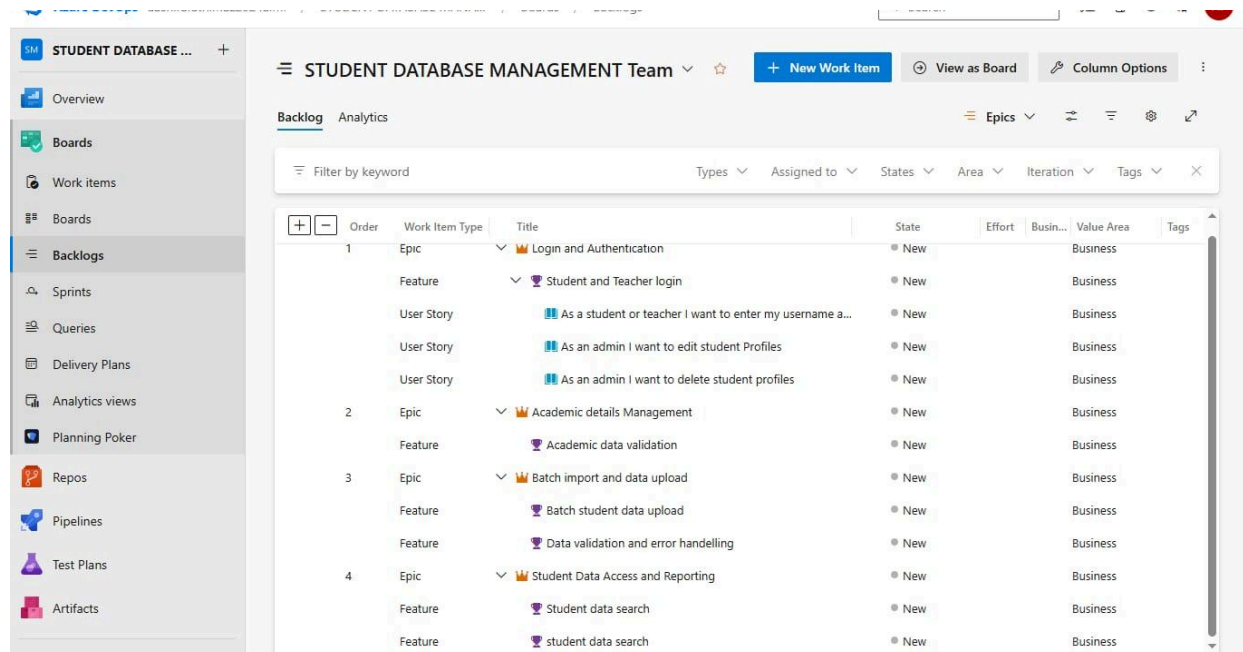
EXP NO: 3
DATE:

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

AIM:-

To learn about how to create epics, user story, features, backlogs for your assigned project.

Create Epic, Features, User Stories, Task



1.Fill in Epics

2.Fill in Features

3.Fill in User Story Details

RESULT:-

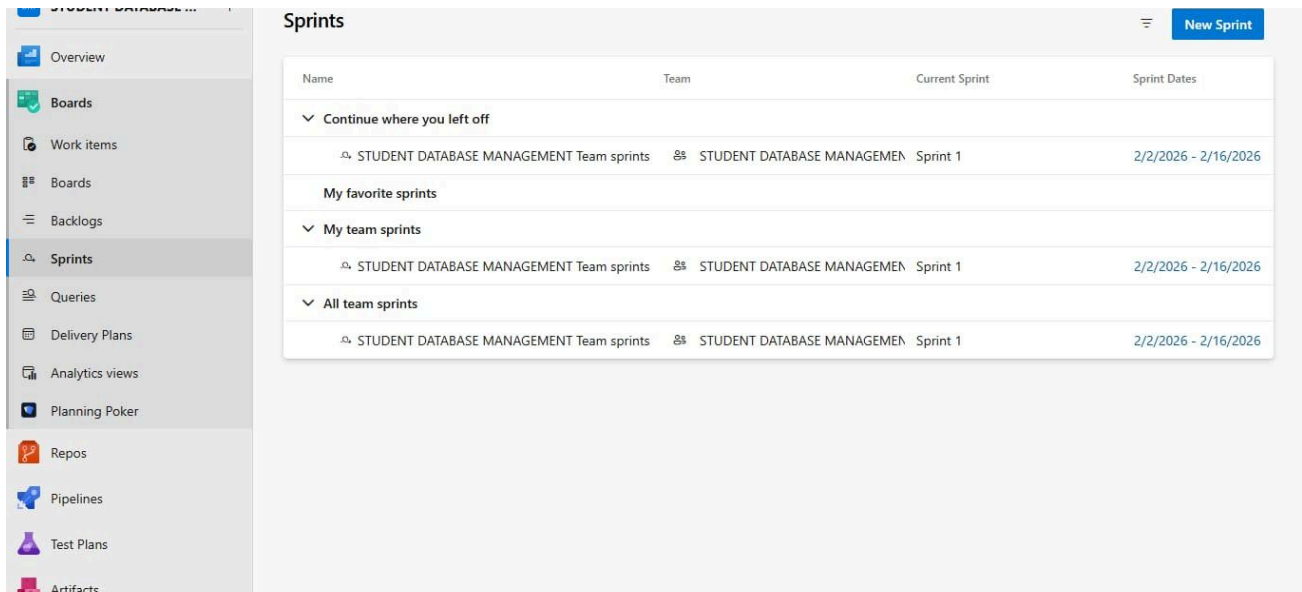
Thus, the creation of epics, features, user story and task has been created successfully.

EXP NO: 4 DATE:	SPRINT PLANNING
----------------------------------	------------------------

AIM:-

To assign user story to specific sprint for the Student Management System Project.

Sprint Planning:-



The screenshot shows the Jira Sprints interface. On the left is a sidebar with navigation options: Overview, Boards, Work items, Backlogs, Sprints (selected), Queries, Delivery Plans, Analytics views, Planning Poker, Repos, Pipelines, Test Plans, and Artifacts. The main area is titled 'Sprints' and features a 'New Sprint' button. Below the title is a table with columns: Name, Team, Current Sprint, and Sprint Dates. The table contains three sections: 'Continue where you left off', 'My favorite sprints', and 'My team sprints'. Each section lists a sprint named 'STUDENT DATABASE MANAGEMENT Team sprints' (with a truncated team name) for 'STUDENT DATABASE MANAGEMENTEN Sprint 1', with dates '2/2/2026 - 2/16/2026'.

Name	Team	Current Sprint	Sprint Dates
Continue where you left off			
STUDENT DATABASE MANAGEMENT Team sprints	STUDENT DATABASE MANAGEMENTEN	Sprint 1	2/2/2026 - 2/16/2026
My favorite sprints			
My team sprints			
STUDENT DATABASE MANAGEMENT Team sprints	STUDENT DATABASE MANAGEMENTEN	Sprint 1	2/2/2026 - 2/16/2026
All team sprints			
STUDENT DATABASE MANAGEMENT Team sprints	STUDENT DATABASE MANAGEMENTEN	Sprint 1	2/2/2026 - 2/16/2026

RESULT:-

The Sprints are created for the Student Management System.

EXP NO: 5
DATE:

POKER ESTIMATION

AIM:-

Create Poker Estimation for the user stories -Student management System.

Poker Estimation:-

STUDENT DATABASE MANA...

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Planning Poker

Repos

Pipelines

Test Plans

Artifacts

Project settings

Poker game, Feb 9, 2026

Current item: As a student or teacher I want to enter my username and password so that I can access my dashboard

Participants: 1 Voted: 3 Share game Become spectator

5 NN

3 AF

1 HR

Type here Save & next Edit vote Reset

Reference items

Game scope

Work item	State	Esti...
As a student or teacher I want to enter my username and pass... USER STORY 3	New	-

USER STORY 3

As a student or teacher I want to enter my username and password so that I can access my dashboard

Unassigned 0 0 1

State New Iteration STUDENT DATABAS...

Description

jhybhybhy

Acceptance Criteria

User name and password has to be valid

RESULT:-

The Estimation/Story Points is created for the project using Poker Estimation.

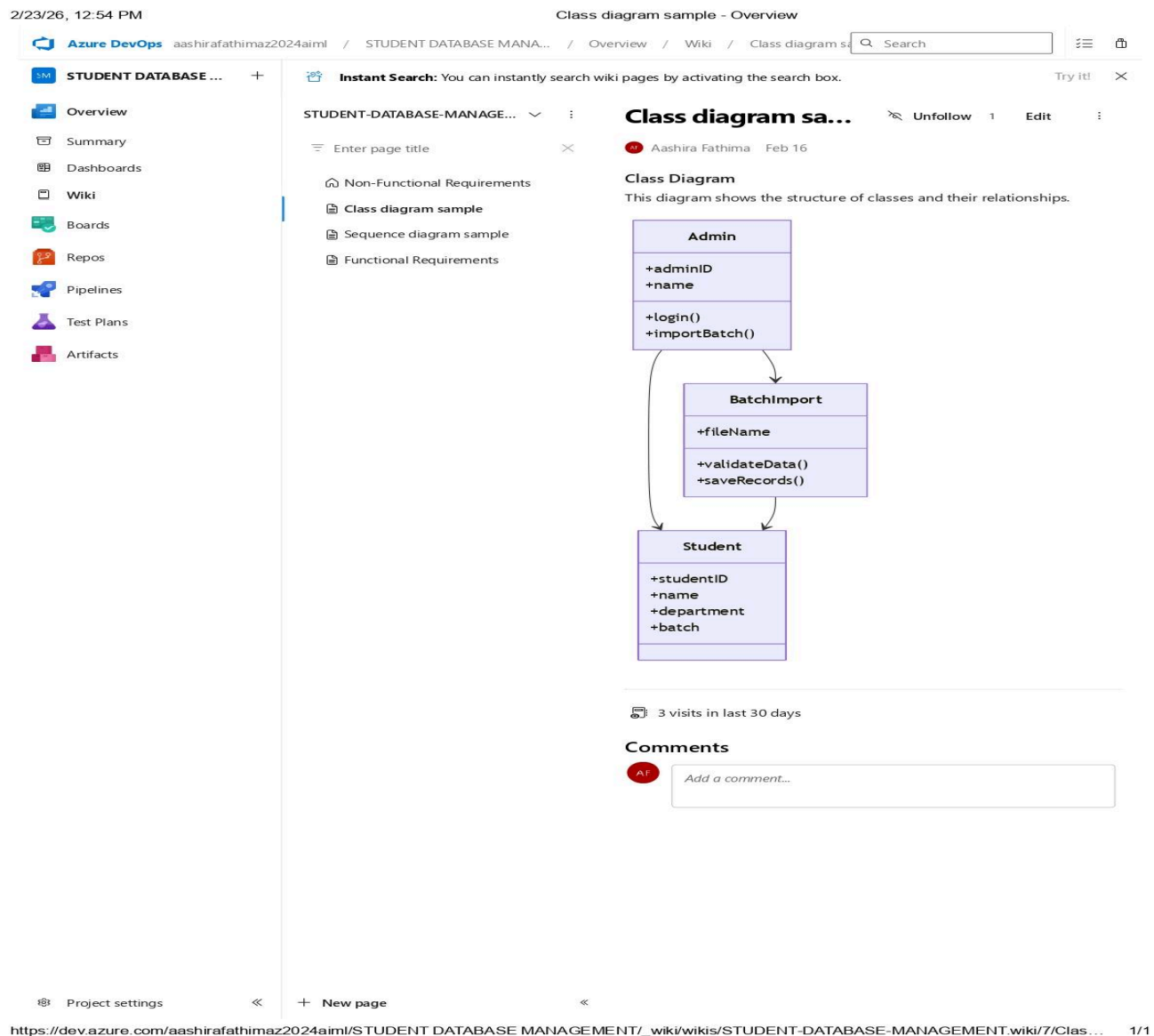
EXP NO: 6
DATE:

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

AIM:-

To Design a Class Diagram and Sequence Diagram for the given Project.

6A. CLASS DIAGRAM:-



6B. SEQUENCE DIAGRAM:-

STUDENT DATABASE MANA...

+

Overview

Summary

Dashboards

Wiki

Boards

Repos

Pipelines

Test Plans

Artifacts

Instant Search: You can instantly search wiki pages by activating the search box.

Try it!

STUDENT-DATABASE-MANA...

Enter page title

Non-Functional Requirements

Class diagram sample

Sequence diagram sample

Functional Requirements

Sequence diagram sample

Aashira Fathima Feb 16

Use case Diagram

This diagram shows how Admin interacts with the system features.

```
graph LR; Admin --> Login; Admin --> AddStudent; Admin --> EditStudent; Admin --> DeleteStudent; Admin --> BatchImport; Admin --> GenerateReport; BatchImport --> ValidateData; GenerateReport --> ExportReport;
```

Sequence Diagram

This diagram shows the interaction flow between Admin and system components.

```
sequenceDiagram; participant Admin; participant System; participant Database; participant ReportModule; Admin->>System: Login; System-->>Admin: Authentication Success; Admin->>System: Upload Student File; System->>Database: Validate Data; Database-->>System: Save Valid records; System-->>Admin: Confirmation; Admin->>System: Generate Report; System->>ReportModule: Create Report; ReportModule-->>System: Export Report;
```

2 visits in last 30 days

Comments

Af

Add a comment...

https://dev.azure.com/aashirafathimaz2024aiml/STUDENT DATABASE MANAGEMENT/_wiki/wikis/STUDENT-DATABASE-MANAGEMENT.wiki/5/Seq... 1/1

Result:

TheClass Diagram and Sequence Diagram is designed Successfully for given project.

EXP NO: 7
DATE:

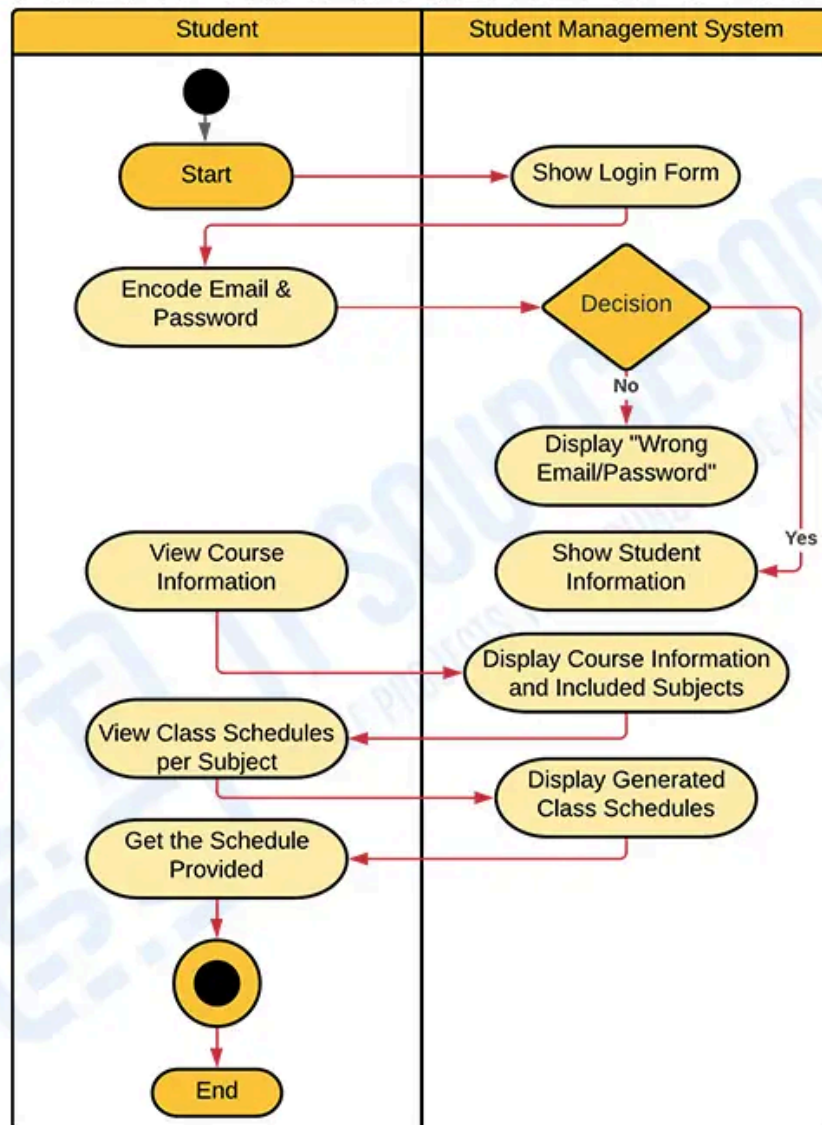
**DESIGNING USE CASE DIAGRAMS AND
ACTIVITY DIAGRAM FOR PROJECT
ARCHITECTURE**

AIM:-

To Design an activity Diagram and use case Diagram for the given Project.

7A. Activity Diagram:-

STUDENT MANAGEMENT SYSTEM



ACTIVITY DIAGRAM

7B. USE CASE DIAGRAM:-

2/23/26, 12:53 PM

Sequence diagram sample - Overview

Azure DevOps aashirafathimaz2024aiml / STUDENT DATABASE MANA... / Overview / Wiki / Sequence diagram sample

STUDENT DATABASE MANA... +

Instant Search: You can instantly search wiki pages by activating the search box.

Try it!

STUDENT-DATABASE-MANAGE...
Enter page title

- Non-Functional Requirements
- Class diagram sample
- Sequence diagram sample
- Functional Requirements

Aashira Fathima Feb 16

Use case Diagram
This diagram shows how Admin interacts with the system features.

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graph LR; Admin --> Login; Admin --> AddStudent; Admin --> EditStudent; Admin --> DeleteStudent; Admin --> BatchImport; Admin --> GenerateReport; BatchImport --> ValidateData; GenerateReport --> ExportReport;
```

Sequence Diagram
This diagram shows the interaction flow between Admin and system components.

```
sequenceDiagram; participant Admin; participant System; participant Database; participant ReportModule; Admin->>System: Login; System-->>Admin: Authentication Success; Admin->>System: Upload Student File; System->>Database: Validate Data; Database-->>System: Save Valid Records; System-->>Admin: Confirmation; Admin->>System: Generate Report; System->>ReportModule: Create Report; ReportModule-->>System: Export Report; System-->>Admin;
```

2 visits in last 30 days

Comments

Add a comment...

Project settings New page

https://dev.azure.com/aashirafathimaz2024aiml/STUDENT DATABASE MANAGEMENT/_wiki/wikis/STUDENT-DATABASE-MANAGEMENT.wiki/5/Seq...

1/1

RESULT:-

Thus activity and use case diagram has been designed successfully for Student Management System

EXP NO: 8 DATE:	TESTING – TEST PLANS AND TEST CASES
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AIM:-

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case **Test Case Design Procedure**

1. Understand Core Features of the Application

- User Signup & Login
- Student Enrollment in Courses
- Assignment Submission & Evaluation
- Viewing Grades and Attendance
- Admin Role Management and Report Generation

2. Define User Interactions

- Each test case simulates a real user behavior (e.g., logging in, submitting an assignment, viewing results).

3. Design Happy Path Test Cases

- Focused on validating that all core functionalities work correctly under normal conditions.
- **Example:** Student registers and logs in, enrolls in a course, submits assignment, and views grades.

4. Design Error Path Test Cases

- Simulate invalid inputs, system issues, or failed actions to ensure proper error handling.
- **Example:** Login with wrong credentials, submission without attachment, unauthorized access to admin panel.

5. Break Down Steps and Expected Results

- Each test case includes a clear sequence of actions and expected results.
- Ensures both manual testers and automation tools can follow the process easily.

6. Use Clear Naming and IDs

- Test cases are uniquely identified (e.g., TC01 – Student Login Success, TC12 – Invalid Assignment Submission).
- Facilitates easy mapping to features and tracking in Azure DevOps.

7. Separate Test Suites

- Grouped by functionality such as:
 - Login and Registration

- Course Enrollment
- Assignment Submission
- Report Generation
- Admin Functions
- Improves organization and enables focused execution in Azure DevOps.

8. Prioritize and Review

- High-priority assigned to critical workflows like login, course access, and grading.
- Reviewed for completeness, accuracy, and alignment with user stories and feature definitions.

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Student Management System– Test Plans

USER STORIES:-

- As a student, I want to sign up and log in securely so I can access academic services. (ID: 101)
- As a student, I need to enroll in courses to participate in classes. (ID: 102)
- As a student, I should be able to submit assignments online for evaluation. (ID: 103)
- As a student, I want to view grades and attendance in one place. (ID: 104)
- As an admin, I want to manage users and generate performance reports. (ID: 105)

Test Suites:-

Test suit : TS01 – User Login & Registration (ID: 201)

1. TC01 – Successful Student Sign-Up

- **Action:**
 - Go to the Sign-Up page
 - Enter valid name, email, and password
 - Click "Sign Up"
- **Expected Result:**
 - Student account is created and redirected to dashboard
- **Type:** Happy Path

2. TC02 – Login with Valid Credentials

- **Action:**
 - Go to Login page
 - Enter valid student email & password
 - Click "Login"
- **Expected Result:**
 - Logged in and taken to student dashboard
- **Type:** Happy Path

3. TC03 – Sign-Up with Existing Email

- **Action:**

- Go to Sign-Up
- Enter already registered email
- Click "Sign Up"

• **Expected Result:**

- Error message: "Email already registered"

• **Type:** Error Path

4. TC04 – Login with Incorrect Password

• **Action:**

- Enter wrong password
- Click "Login"

• **Expected Result:**

- Error: "Invalid username or password"

• **Type:** Error Path

Test Suite: TS02 – Course Enrollment (ID: 202)

1. TC05 – Enroll in Available Course

• **Action:**

- Login → Navigate to Course List
- Select a course → Click "Enroll"

• **Expected Result:**

- Enrollment confirmation shown

• **Type:** Happy Path

2. TC06 – Try Enrolling in a Full Course

• **Action:**

- Select a course with no seats available
- Click "Enroll"

• **Expected Result:**

- Error: "Enrollment failed – course full"

• **Type:** Error Path

Test Suite: TS03 – Assignment Submission (ID: 203)

1. TC07 – Submit Assignment Successfully

• **Action:**

- Login → Go to Assignments
- Upload file and click "Submit"

• **Expected Result:**

- Submission confirmation message shown

• **Type:** Happy Path

2. TC08 – Submit Without Uploading File

• **Action:**

- Open assignment submission

- Click "Submit" without file

• **Expected Result:**

- Error: "Please upload a file before

submitting" • **Type:** Error Path

Test Suite: TS04 – View Grades & Attendance (ID: 204)

1. TC09 – View Grades Dashboard

• **Action:**

- Login → Go to "My Grades"

• **Expected Result:**

- All grades are displayed by subject

• **Type:** Happy Path

2. TC10 – Fail to Fetch Grades

(Offline Scenario) • Action:

- Disconnect network → Go to "My Grades"

• **Expected Result:**

- Error: "Unable to fetch data – check

connection" • **Type:** Error Path

Test Suite: TS05 – Admin Management (ID: 205)

1. TC11 – Assign Role to New User

• **Action:**

- Login as Admin → Navigate to User Management
- Select user and assign role

• **Expected Result:**

- Confirmation: "Role assigned successfully"

• **Type:** Happy Path

2. TC12 – Generate Report with No Data

• **Action:**

- Navigate to Reports → Choose empty semester
- Click "Generate"

• **Expected Result:**

- Message: "No data available to

generate report" • **Type:** Error Path

3. Test Cases

4. Installation of test

5. Running the test cases
6. Recording the test case
7. Creating the bug

Test Results

shirafathimaz2024aiml / STUDENT DATABASE MANA... / Test Plans / Attendance management

Search

SE ...

Runner - Test Plans - Personal - Mic...

https://dev.azure.com/aashirafathimaz20...

18*: TC-01 Login with valid credentials

- Open login page
EXPECTED RESULT
User is redirected to dashboard
- Enter valid username
- Enter valid password
COMMENT
invalid password entered
- Click Login
COMMENT
Prev step failed

Attendance management (ID: 17)

Define Execute Chart

Test Points (3 items)

Title	Outcome
☒ TC-01 Login with valid credentials	Failed
☐ TC-02 Login with invalid credentials	Passed
☐ TC-03 Login with empty username	Active

Run for web

8. Test report summary

STUDENT DATABASE ...

Recently updated Back to Work Items

5 of 19

BUG 21

21 BG-01 Invalid password entered

Navdeep REC 0 Comments Add Tag

Save Follow

Updated by Aashira Fathima: 11h ago

State: Active Area: STUDENT DATABASE MANAGEMENT

Reason: Approved Iteration: STUDENT DATABASE MANAGEMENT\Sprint 1

Details

Repro Steps

3/23/2026 7:03 AM Bug filed on "TC-01 Login with valid credentials"

Step no.	Result	Title
1.	Passed	Open login page
		Expected Result User is redirected to dashboard
2.	Passed	Enter valid username
3.	None	Enter valid password
4.	None	Click Login

Planning

Resolved Reason

Story Points

Priority

2

Severity

3 - Medium

Activity

Effort (Hours)

Original Estimate

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

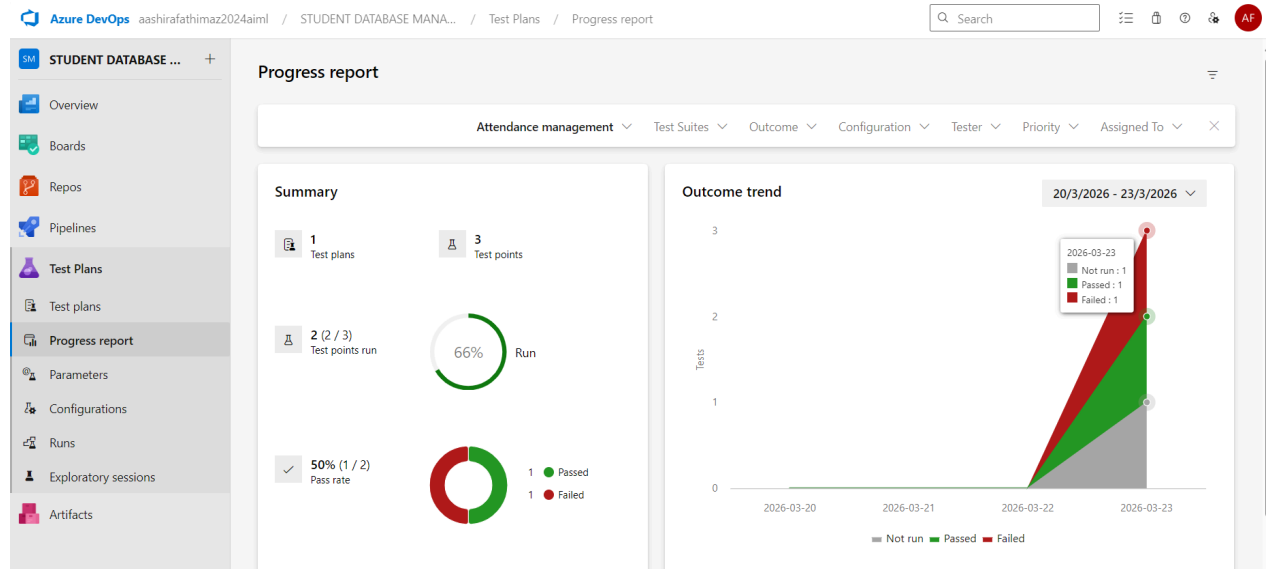
Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started

1. Assigning bug to the developer and changing state

9. Progress report



Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

EXP NO: 9 DATE:	CI/CD PIPELINES IN AZURE
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AIM:

To implement a Continuous Integration and Continuous Deployment (CI/CD) pipeline in Azure DevOps for automating the build, testing, and deployment process of the Student Management System, ensuring faster delivery and improved software quality.

PROCEDURE:

Steps to Create and implement pipelines in Azure:

1. Sign in to Azure DevOps and Navigate to Your Project
Log in to dev.azure.com, select your organization, and open the project where your Student Management System code resides.
2. Connect a Code Repository (Azure Repos or GitHub)
Ensure your application code is stored in a Git-based repository such as Azure Repos or GitHub. This will be the source for triggering builds and deployments in your pipeline.
3. Create a New Pipeline
Go to the Pipelines section on the left panel and click “Create Pipeline”.
Choose your source (e.g., Azure Repos Git or GitHub), and then select the repository containing your project code.
4. Choose the Pipeline Configuration
You can select either the YAML-based pipeline (recommended for version control and automation) or the Classic Editor for a GUI-based setup.
If using YAML, Azure DevOps will suggest a template or allow you to define your own.
5. Define Build Stage (CI - Continuous Integration) from YAML file
6. Install dependencies (e.g., npm install, dotnet restore)
7. Build the application (dotnet build, npm run build)
8. Run unit tests (dotnet test, npm test)
9. Publish build artifacts to be used in the release stage
10. Save and Run the Pipeline for the First Time
Save the YAML or build definition and click “Run”.

Azure will fetch the latest code and execute the defined build and test stages.

11. Configure Continuous Deployment (CD)

Navigate to the Releases tab under Pipelines and click “New Release Pipeline”.

Add an Artifact (from the build stage) and create a new Stage (e.g., Development, Production).

12. Configure the CD stage with deployment tasks such as deploying to Azure App Service, running database migrations or scripts, and restarting services using the Azure App Service Deploy task linked to your subscription and app details.

13. Set Triggers and Approvals

Enable continuous deployment trigger so the release pipeline runs automatically after a successful build.

For production environments, configure pre-deployment approvals to ensure manual verification before release.

14. Monitor Pipelines and Manage Logs

View all pipeline runs under the Runs section.

Check logs for build/test/deploy stage to debug any errors.

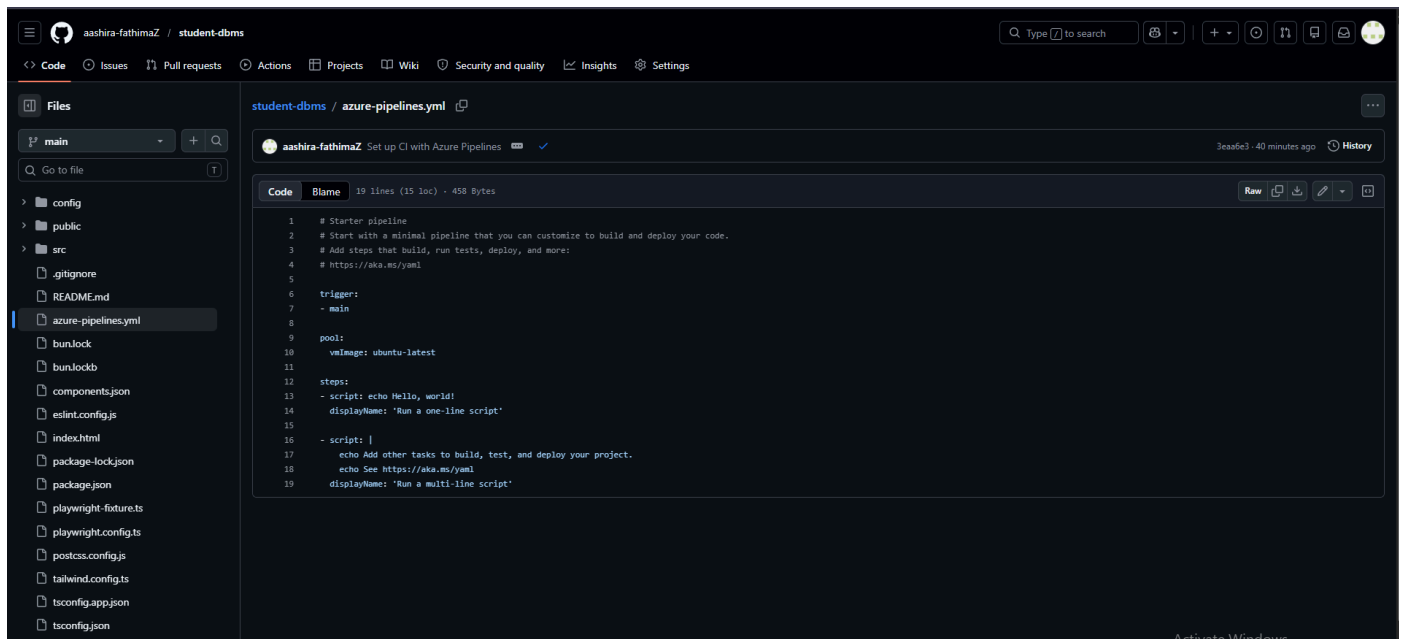
You can also integrate email alerts or Microsoft Teams notifications for build failures.

15. Review and Maintain Pipelines

Regularly update your pipeline tasks or YAML configurations as your application grows. Ensure pipeline runs are clean and artifacts are stored securely.

Integrate quality gates and code coverage policies to maintain code quality.

The screenshot displays the Azure DevOps web interface. The top navigation bar shows the user 'aashirafathimaz2024aiml' and the current path: 'STUDENT DATABASE MANA...' / Pipelines / aashira-fathimaZ.student-d... / 20260504.1. A search bar and utility icons are on the right. The left sidebar contains a menu with 'STUDENT DATABASE ...', 'Overview', 'Boards', 'Repos', 'Pipelines', 'Environments', 'Library', 'Test Plans', and 'Artifacts'. The main content area is titled '#20260504.1 • Set up CI with Azure Pipelines' and includes a 'Run new' button. A message states: 'This run is being retained as one of 3 recent runs by pipeline.' Below this, the 'Summary' tab is active, showing 'Individual CI by aashira-fathimaZ'. It lists the repository 'aashira-fathimaZ/student-dbms', branch 'main', commit '3eaa6e36', and shows the run started 'Today at 12:30 PM' and elapsed '5s'. It also indicates '0 work items' and '0 artifacts'. A 'View change' button is present. The 'Stages' tab shows a single stage 'Stage' with '1 job completed' in '6s'. A 'Rerun Stage' button is at the bottom.



RESULT:

Thus the pipelines for the given project “Student Management System has been executed successfully

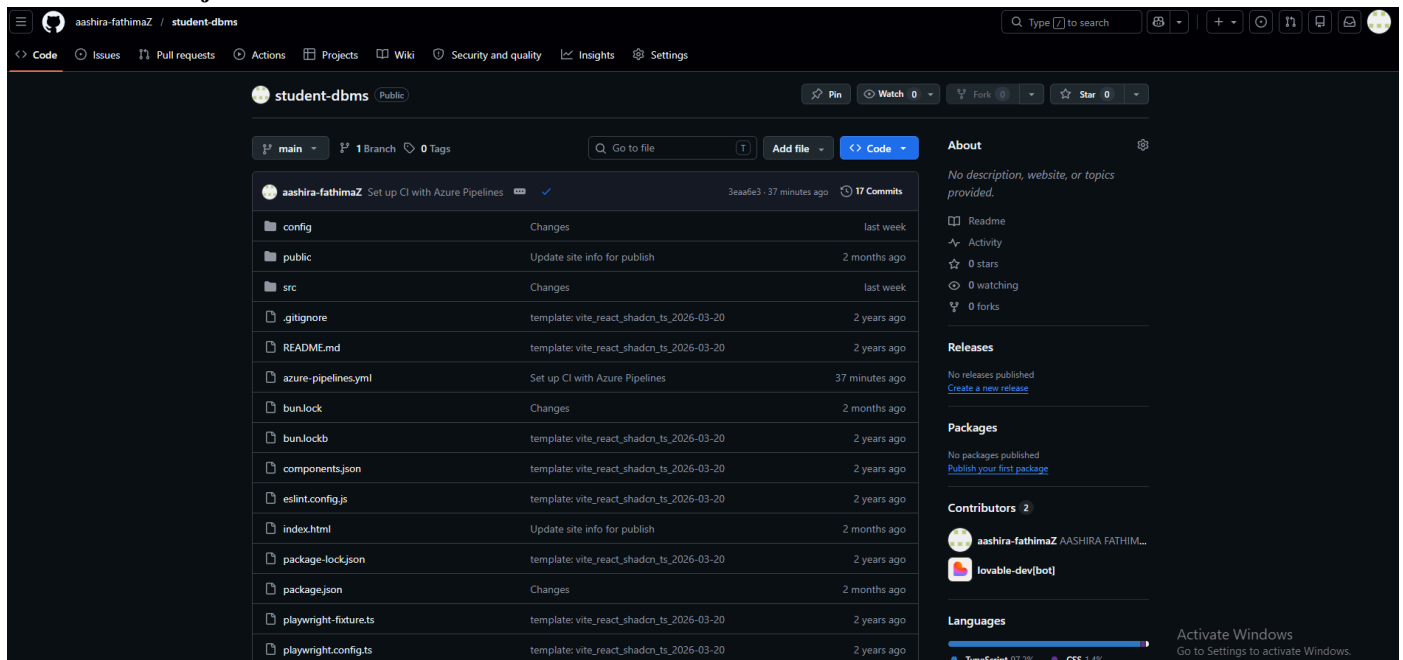
EXP NO: 10
DATE:

GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Student Management System project.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.

